

Week 10: Temporal data

20/03/26

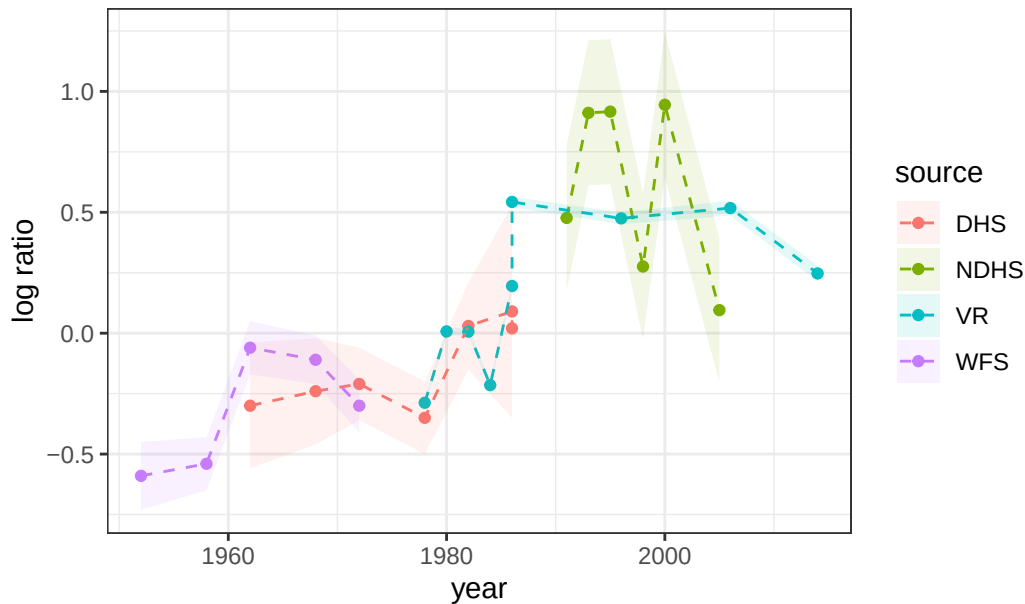
Child mortality in Sri Lanka

In this lab you will be fitting a couple of different models to the data about child mortality in Sri Lanka, which was used in the lecture. Here's the data and the plot from the lecture:

```
library(tidyverse)
library(here)
library(rstan)
library(tidybayes)

lka <- read_csv(here("data/lka.csv"))
ggplot(lka, aes(year, logit_ratio)) +
  geom_point(aes( color = source)) +
  geom_line(aes( color = source), lty = 2) +
  geom_ribbon(aes(ymin = logit_ratio - se,
                 ymax = logit_ratio + se,
                 fill = source), alpha = 0.1) +
  theme_bw()+
  labs(title = "Ratio of neonatal to other child mortality (logged), Sri Lanka", y = "log ra
```

Ratio of neonatal to other child mortality (logged), Sri Lanka



Fitting a linear model

Let's firstly fit a linear model in time to these data. Here's the code to do this:

```
observed_years <- lka$year
years <- min(observed_years):max(observed_years)
nyears <- length(years)

stan_data <- list(y = lka$logit_ratio, year_i = observed_years - years[1]+1,
  T = nyears, years = years, N = length(observed_years),
  mid_year = mean(years), se = lka$se)

mod <- stan(data = stan_data,
  file = here("code/models/lka_linear_me.stan"))
```

```
Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: 'Apple clang version 12.0.0 (clang-1200.0.31.1)'
using SDK: 'MacOSX11.0.sdk'
clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/Rcpp/include/"
I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/"
I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/un
```

```

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/BH/include" -
I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/StanHeaders/include/
I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppParallel/include/
I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/rstan/include" -
DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -
DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include '/Library/Frameworks/R.framework/Versions/4.5-
x86_64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp' -
D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/x86_64/include -fPIC -
falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/
In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/
In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/
/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
    ~~~~~~
1 error generated.
make: *** [foo.o] Error 1

```

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 2.1e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.21 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1: Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1: Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1: Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1: Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1: Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 0.04 seconds (Warm-up)
Chain 1:                0.035 seconds (Sampling)

```

Chain 1: 0.075 seconds (Total)

Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

Chain 2:

Chain 2: Gradient evaluation took 8e-06 seconds

Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.08 seconds.

Chain 2: Adjust your expectations accordingly!

Chain 2:

Chain 2:

Chain 2: Iteration: 1 / 2000 [0%] (Warmup)

Chain 2: Iteration: 200 / 2000 [10%] (Warmup)

Chain 2: Iteration: 400 / 2000 [20%] (Warmup)

Chain 2: Iteration: 600 / 2000 [30%] (Warmup)

Chain 2: Iteration: 800 / 2000 [40%] (Warmup)

Chain 2: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 2: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 2: Iteration: 1200 / 2000 [60%] (Sampling)

Chain 2: Iteration: 1400 / 2000 [70%] (Sampling)

Chain 2: Iteration: 1600 / 2000 [80%] (Sampling)

Chain 2: Iteration: 1800 / 2000 [90%] (Sampling)

Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)

Chain 2:

Chain 2: Elapsed Time: 0.045 seconds (Warm-up)

Chain 2: 0.042 seconds (Sampling)

Chain 2: 0.087 seconds (Total)

Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

Chain 3:

Chain 3: Gradient evaluation took 5e-06 seconds

Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.05 seconds.

Chain 3: Adjust your expectations accordingly!

Chain 3:

Chain 3:

Chain 3: Iteration: 1 / 2000 [0%] (Warmup)

Chain 3: Iteration: 200 / 2000 [10%] (Warmup)

Chain 3: Iteration: 400 / 2000 [20%] (Warmup)

Chain 3: Iteration: 600 / 2000 [30%] (Warmup)

Chain 3: Iteration: 800 / 2000 [40%] (Warmup)

Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)

```
Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.045 seconds (Warm-up)
Chain 3:           0.036 seconds (Sampling)
Chain 3:           0.081 seconds (Total)
Chain 3:
```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

```
Chain 4:
Chain 4: Gradient evaluation took 6e-06 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.06 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 0.044 seconds (Warm-up)
Chain 4:           0.036 seconds (Sampling)
Chain 4:           0.08 seconds (Total)
Chain 4:
```

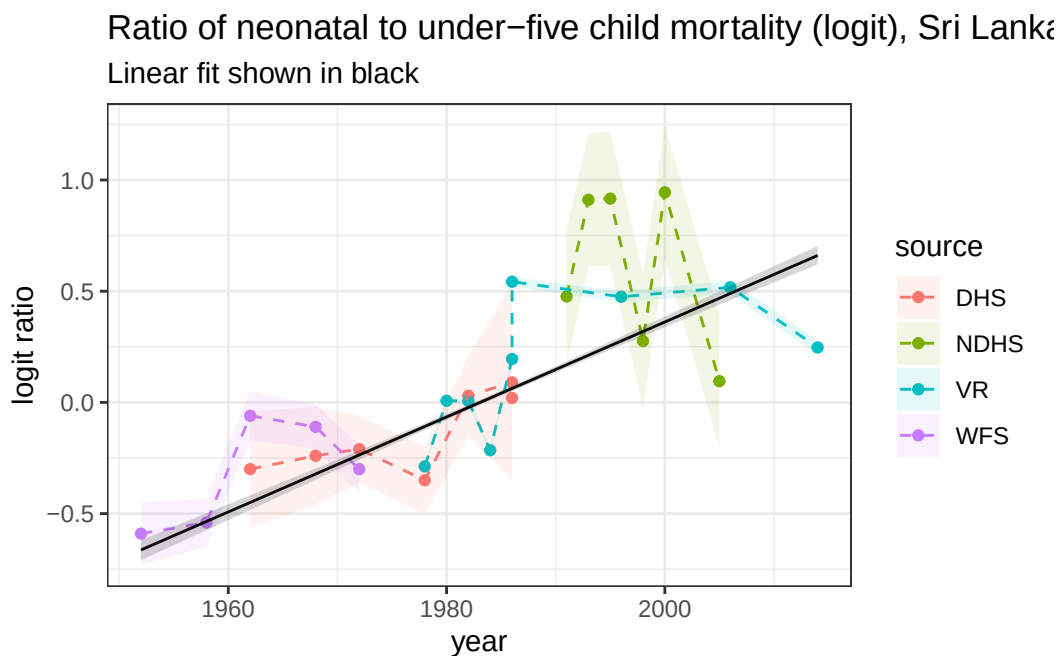
Extract the results:

```
res <- mod |>
  gather_draws(mu[t]) |>
  median_qi() |>
  mutate(year = years[t])
```

Plot the results:

```
ggplot(lka, aes(year, logit_ratio)) +
  geom_point(aes( color = source)) +
  geom_line(aes( color = source), lty = 2) +
  geom_ribbon(aes(ymin = logit_ratio - se,
                 ymax = logit_ratio + se,
                 fill = source), alpha = 0.1) +

  theme_bw()+
  geom_line(data = res, aes(year, .value)) +
  geom_ribbon(data = res, aes(y = .value, ymin = .lower, ymax = .upper), alpha = 0.2)+
  theme_bw()+
  labs(title = "Ratio of neonatal to under-five child mortality (logit), Sri Lanka",
       y = "logit ratio", subtitle = "Linear fit shown in black")
```



Question 1

Project the linear model above out to 2024 by adding a **generated quantities** block in Stan (do the projections based on the expected value μ). Plot the resulting projections on a graph similar to that above.

```
stan_data <- list(y = lka$logit_ratio, year_i = observed_years - years[1]+1,
                 T = nyears, years = years, N = length(observed_years),
                 mid_year = mean(years), se = lka$se,
```

```

years_p = (years[length(years)]+1):2024, P = 10, max_se = max(lka$se))

mod2 <- stan(data = stan_data,
             file = here("code/models/lka_linear_me_proj.stan"))

```

```

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: 'Apple clang version 12.0.0 (clang-1200.0.31.1)'
using SDK: 'MacOSX11.0.sdk'
clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/Rcpp/include/"
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/"
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/un
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/BH/include" -
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/StanHeaders/include/"
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/StanHeaders/include/"
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppParallel/include/"
-I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/rstan/include" -
DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -
DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include '/Library/Frameworks/R.framework/Versions/4.5-
x86_64/Resources/library/StanHeaders/include/Stan/math/prim/fun/Eigen.hpp' -
D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/x86_64/include -fPIC -
falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/Stan/math/prim/fun/Eigen.hpp:1:
In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/Stan/math/prim/fun/Eigen.hpp:1:
In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/Eigen/
/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
^~~~~~
1 error generated.
make: *** [foo.o] Error 1

```

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 2.5e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.25 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [ 0%] (Warmup)
Chain 1: Iteration:   200 / 2000 [ 10%] (Warmup)

```

```

Chain 1: Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 0.051 seconds (Warm-up)
Chain 1: 0.049 seconds (Sampling)
Chain 1: 0.1 seconds (Total)
Chain 1:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

```

Chain 2:
Chain 2: Gradient evaluation took 5e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.05 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2: Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2: Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2: Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2: Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 0.051 seconds (Warm-up)
Chain 2: 0.05 seconds (Sampling)
Chain 2: 0.101 seconds (Total)
Chain 2:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

```

Chain 3:

```


Chain 3: Gradient evaluation took 7e-06 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.07 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration: 1 / 2000 [0%] (Warmup)
Chain 3: Iteration: 200 / 2000 [10%] (Warmup)
Chain 3: Iteration: 400 / 2000 [20%] (Warmup)
Chain 3: Iteration: 600 / 2000 [30%] (Warmup)
Chain 3: Iteration: 800 / 2000 [40%] (Warmup)
Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.051 seconds (Warm-up)
Chain 3: 0.038 seconds (Sampling)
Chain 3: 0.089 seconds (Total)
Chain 3:

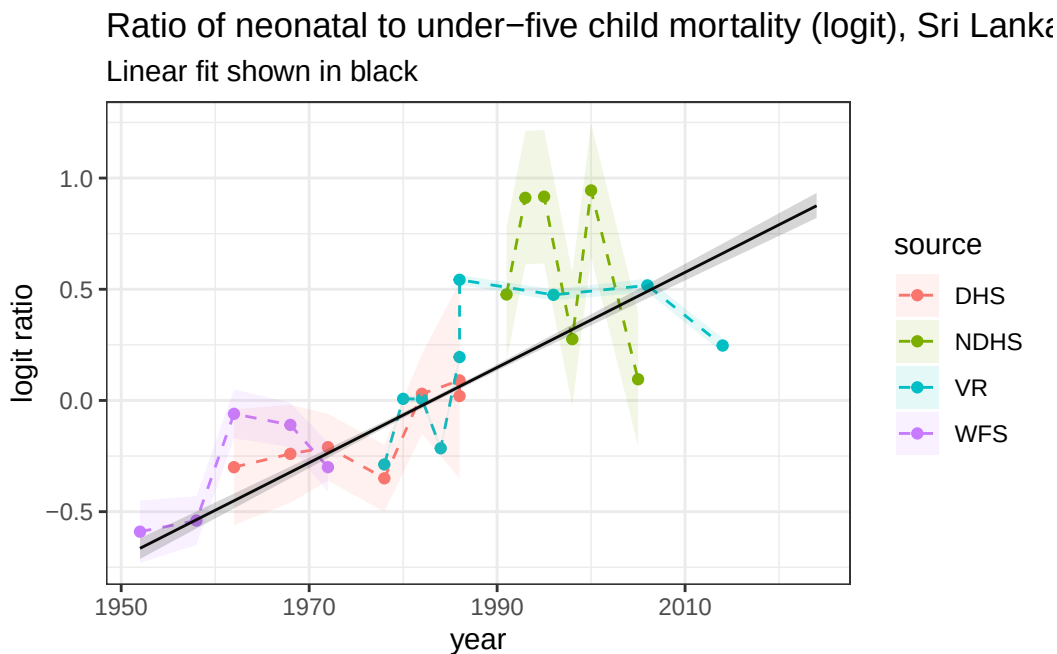
SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

Chain 4:
Chain 4: Gradient evaluation took 5e-06 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.05 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration: 1 / 2000 [0%] (Warmup)
Chain 4: Iteration: 200 / 2000 [10%] (Warmup)
Chain 4: Iteration: 400 / 2000 [20%] (Warmup)
Chain 4: Iteration: 600 / 2000 [30%] (Warmup)
Chain 4: Iteration: 800 / 2000 [40%] (Warmup)
Chain 4: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:

Chain 4: Elapsed Time: 0.041 seconds (Warm-up)
 Chain 4: 0.037 seconds (Sampling)
 Chain 4: 0.078 seconds (Total)
 Chain 4:

```
res2 <- mod2 |>
  gather_draws(mu[t], mu_proj[p]) |>
  median_qi() |>
  mutate(year = ifelse(!is.na(t), years[t], ((years[length(years)]+1):2024)[p]))

ggplot(lka, aes(year, logit_ratio)) +
  geom_point(aes( color = source)) +
  geom_line(aes( color = source), lty = 2) +
  geom_ribbon(aes(ymin = logit_ratio - se,
                 ymax = logit_ratio + se,
                 fill = source), alpha = 0.1) +
  theme_bw()+
  geom_line(data = res2, aes(year, .value)) +
  geom_ribbon(data = res2, aes(y = .value, ymin = .lower, ymax = .upper), alpha = 0.2)+
  theme_bw()+
  labs(title = "Ratio of neonatal to under-five child mortality (logit), Sri Lanka",
       y = "logit ratio", subtitle = "Linear fit shown in black")
```



Question 2

The projections above are for the logit of the ratio of neonatal to under-five child mortality. You can download estimates of the under-five child mortality from 1951 to 2024 here: <https://childmortality.org/all-cause-mortality/data/estimates?refArea=LKA>. Use these data to get estimates and projections of neonatal mortality for Sri Lanka, and plot the results.

Random walks

Question 3

Code up and estimate a first order random walk model to fit to the Sri Lankan data, taking into account measurement error, and project out to 2024.

```
mod3 <- stan(data = stan_data,
             file = here("code/models/lka_rw.stan"))
```

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c

using C compiler: 'Apple clang version 12.0.0 (clang-1200.0.31.1)'

using SDK: 'MacOSX11.0.sdk'

clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -

DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/Rcpp/include"

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/"

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/un

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/BH/include" -

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/StanHeaders/include/

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/StanHeaders/include/

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppParallel/include

I"/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/rstan/include" -

DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -

DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -

D_HAS_AUTO_PTR_ETC=0 -include '/Library/Frameworks/R.framework/Versions/4.5-

x86_64/Resources/library/StanHeaders/include/Stan/math/prim/fun/Eigen.hpp' -

D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/x86_64/include -fPIC -

falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o

In file included from <built-in>:1:

In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/

In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/

In file included from /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/

/Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/library/RcppEigen/include/Eigen

#include <cmath>

~~~~~  
1 error generated.  
make: \*\*\* [foo.o] Error 1

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 1).

Chain 1:

Chain 1: Gradient evaluation took 3e-05 seconds

Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.3 seconds.

Chain 1: Adjust your expectations accordingly!

Chain 1:

Chain 1:

Chain 1: Iteration: 1 / 2000 [ 0%] (Warmup)

Chain 1: Iteration: 200 / 2000 [ 10%] (Warmup)

Chain 1: Iteration: 400 / 2000 [ 20%] (Warmup)

Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)

Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)

Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)

Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)

Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)

Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)

Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)

Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)

Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)

Chain 1:

Chain 1: Elapsed Time: 0.245 seconds (Warm-up)

Chain 1: 0.203 seconds (Sampling)

Chain 1: 0.448 seconds (Total)

Chain 1:

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 2).

Chain 2:

Chain 2: Gradient evaluation took 8e-06 seconds

Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.08 seconds.

Chain 2: Adjust your expectations accordingly!

Chain 2:

Chain 2:

Chain 2: Iteration: 1 / 2000 [ 0%] (Warmup)

Chain 2: Iteration: 200 / 2000 [ 10%] (Warmup)

Chain 2: Iteration: 400 / 2000 [ 20%] (Warmup)

Chain 2: Iteration: 600 / 2000 [ 30%] (Warmup)

Chain 2: Iteration: 800 / 2000 [ 40%] (Warmup)

Chain 2: Iteration: 1000 / 2000 [ 50%] (Warmup)

Chain 2: Iteration: 1001 / 2000 [ 50%] (Sampling)

Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)  
Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)  
Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)  
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)  
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)  
Chain 2:  
Chain 2: Elapsed Time: 0.247 seconds (Warm-up)  
Chain 2: 0.214 seconds (Sampling)  
Chain 2: 0.461 seconds (Total)  
Chain 2:

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 3).

Chain 3:  
Chain 3: Gradient evaluation took 7e-06 seconds  
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.07 seconds.  
Chain 3: Adjust your expectations accordingly!  
Chain 3:  
Chain 3:  
Chain 3: Iteration: 1 / 2000 [ 0%] (Warmup)  
Chain 3: Iteration: 200 / 2000 [ 10%] (Warmup)  
Chain 3: Iteration: 400 / 2000 [ 20%] (Warmup)  
Chain 3: Iteration: 600 / 2000 [ 30%] (Warmup)  
Chain 3: Iteration: 800 / 2000 [ 40%] (Warmup)  
Chain 3: Iteration: 1000 / 2000 [ 50%] (Warmup)  
Chain 3: Iteration: 1001 / 2000 [ 50%] (Sampling)  
Chain 3: Iteration: 1200 / 2000 [ 60%] (Sampling)  
Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)  
Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)  
Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)  
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)  
Chain 3:  
Chain 3: Elapsed Time: 0.241 seconds (Warm-up)  
Chain 3: 0.205 seconds (Sampling)  
Chain 3: 0.446 seconds (Total)  
Chain 3:

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 4).

Chain 4:  
Chain 4: Gradient evaluation took 8e-06 seconds  
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.08 seconds.  
Chain 4: Adjust your expectations accordingly!  
Chain 4:  
Chain 4:

```

Chain 4: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4: Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4: Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4: Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4: Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4: Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 4: Iteration:  2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 0.259 seconds (Warm-up)
Chain 4:                0.199 seconds (Sampling)
Chain 4:                0.458 seconds (Total)
Chain 4:

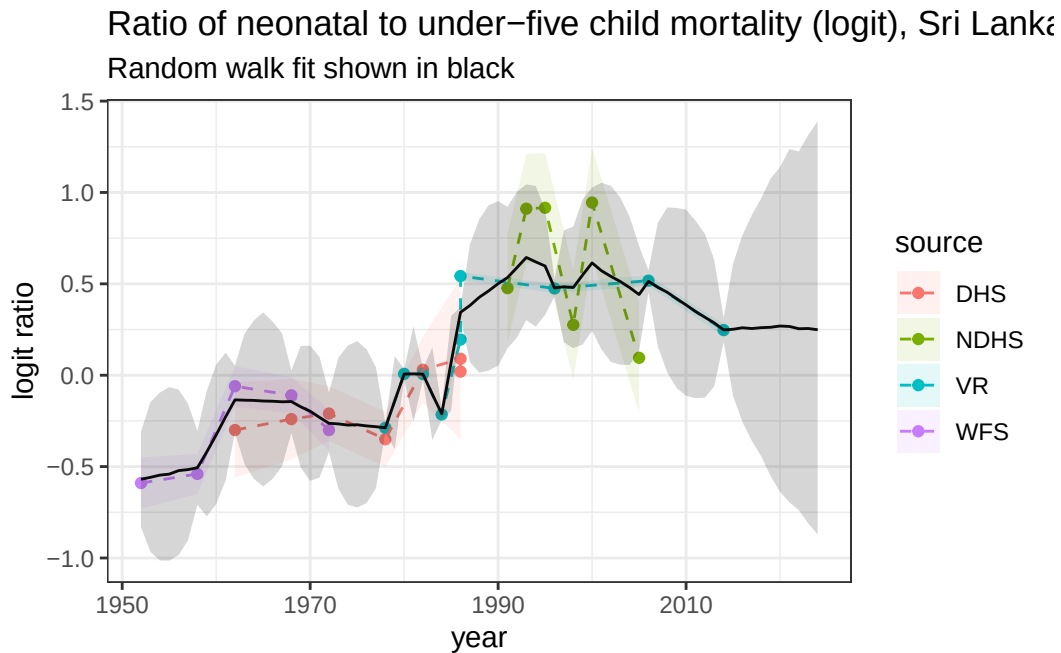
```

```

res3 <- mod3 |>
  gather_draws(alpha[t], alpha_proj[p]) |>
  median_qi() |>
  mutate(year = ifelse(!is.na(t), years[t], ((years[length(years)]+1):2024)[p]))

ggplot(lka, aes(year, logit_ratio)) +
  geom_point(aes( color = source)) +
  geom_line(aes( color = source), lty = 2) +
  geom_ribbon(aes(ymin = logit_ratio - se,
                 ymax = logit_ratio + se,
                 fill = source), alpha = 0.1) +
  theme_bw()+
  geom_line(data = res3, aes(year, .value)) +
  geom_ribbon(data = res3, aes(y = .value, ymin = .lower, ymax = .upper), alpha = 0.2)+
  theme_bw()+
  labs(title = "Ratio of neonatal to under-five child mortality (logit), Sri Lanka",
       y = "logit ratio", subtitle = "Random walk fit shown in black")

```



#### Question 4

Now alter your model above to estimate and project a second-order random walk model (RW2).

#### Question 5

Compare these estimates with the linear fit by plotting everything on the same graph.

#### Question 6

Briefly comment on which model you think is most appropriate, or an alternative model that would be more appropriate in this context.